

Question 1

- (a) Rationalize the closest oxygen atom to oxygen atom distances found in the following compounds:

O_2PtF_6 112 pm

O_2 121 pm

KO_2 133 pm

Na_2O_2 149 pm

Predict which of these species will be paramagnetic. [7]

- (b) Discuss the following observations:

(i) N_2^+ has a longer bond length than N_2 by about 3 pm, but O_2^+ has a shorter bond length than O_2 by 9 pm.

(ii) The N–O bond length in nitrogen monoxide is 115 pm, but in the compound $\text{NO}[\text{ClO}_4]$ it is 106 pm. [5]

- (c) Use the following data:–



(i) to obtain O–O and O–H single bond enthalpy terms. [4]

(ii) to estimate a value for the electronegativity of oxygen on the Pauling scale, where that of hydrogen is 2.2. [3]

Question 2

- (a) Draw and label a molecular orbital energy level diagram for dinitrogen, N_2 .
Annotate it with pictorial representations of the molecular orbitals involved. [8]
- (b) What is the bond order of N_2 ?
How is this reflected in the reactivity and physical properties of N_2 molecules? [4]
- (c) Draw a molecular orbital energy level diagram for CO. How does it differ from that for N_2 ? [5]
- (d) What is the bond order of CO?
Draw a valence-bond (Lewis) representation of the CO molecule.
What is the direction of the dipole in CO? Explain your answer.
The bond dissociation enthalpy of N_2 is 942 kJ mol^{-1} and that for CO is 1072 kJ mol^{-1} . Explain this observation. [5]
- (e) What properties in contrast to those of CO do you predict for NO from its electronic configuration and molecular orbital energy level diagram? [3]